

Endoscopic Image-Based Deep Learning Approach for Predicting Gastrointestinal Diseases

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Abstract— Gastrointestinal diseases such as colorectal cancer, ulcers, and polyps, normal Z-line are continued to be a big worldwide health problems and there must be a necessary precise of diseases and early detection for successful treatment. Endoscopy serves as the standard diagnostic technique. But, its accuracy may compromise the degree of clinicians, fatigue, and subjective checks, which may additionally result in misdiagnoses. To overcome these critical and most improper situations, there should be an implementation of artificial intelligence which have a computer aided diagnosis structures in scientific imaging to have a successful treatment with more accuracy. In the scientific imaging we have a deep learning framework utilizing convolutional neural networks that has been evolved quickly and been tested for the automatic prediction of GI diseases based on endoscopic images using the respective algorithm called CNNs. In this paper we specific, three models such as ResNet50, MobileNetV2, and VGG16 which are used to improve the feature extraction, classification and prediction more efficiently. This paper proposes that ResNet50 model have been designed like CAD that will target to deliver unique, unbiased, and replicable diagnostic assistance, ultimately enhancing medical decision- making and affected person results.

Keywords — Deep Learning, Endoscopy, Artificial intelligence, Computer-Aided Diagnosis, Convolutional Neural Network, Disease Classification.

I. INTRODUCTION

Gastrointestinal diseases constitute an extensive public health and challenge them globally, where there impact are of hundreds and thousands every year and has large mortality and healthcare costs which is a much challenged problem to both doctors and patients to overcome. The specific diagnosis of those situations is critical to implementing the proper remedy and accomplishing higher fitness effects. Presently, endoscopy stands as the touchiest approach to be had for assessing abnormalities within the digestive tract, as it allows physicians to without delay visualize the mucosal layer and regularly permits on the spot therapeutic interventions. Nevertheless, the accuracy of endoscopic interpretation substantially is predicated on the health practitioner's enjoy, the technical sophistication of the manner, and the capability to perceive subtle or complex lesions. Human errors, exhaustion, and versions in clinical judgment can bring about delayed or neglected diagnoses, posing risks to the affected person's fitness. In current years, improvements in artificial intelligence, especially deep gaining knowledge of, have opened new opportunities for computer assisted diagnosis in endoscopy, according to the international agency for research on cancer (IARC) from the World Health Organization, [1]. Convolutional neural networks have verified to be a quite efficient in image classification, making them rather applicable for reading clinical pix. In the context of endoscopic assessments, AI-driven structures can systematically and as it should be assessment huge volumes of imaging statistics, thereby decreasing dependence on human interpretation. This study focuses on the creation and demonstration of a fully automatic diagnostic framework based totally on deep gaining knowledge of that goals to differentiate between not unusual GI diseases using endoscopic images graphs, by integrating AI into medical exercise, the lengthy-time.

In the period of modern instances, improvement in artificial intelligence (AI), especially in deep gaining knowledge of, has created new opportunities for computer aided diagnosis in endoscopic processes. Convolutional neural networks have given some awesome effectiveness in images graph which to be in quality for the evaluation of scientific imaging. When carried out in endoscopic practices, AI- powered systems can evaluate giant portions of image information with high accuracy, consistency, and impartiality, reducing dependency on subjective interpretations via humans. This research interests to create and validate a totally computerized framework based totally mostly on deep learning for classifying often occurring gastrointestinal (GI) situations from endoscopic images.

II. RELATED WORK

In recent times, deep learning and data vision techniques have proven to be effective tools within the framework of image analysis. Several studies have examined their use to identify and classify gastrointestinal (GI) diseases through endoscopic images. The emergence of benchmark data sets, nervous network architecture progression and implementation of transmission learning techniques have greatly increased clinical accuracy. This sector undergoes with some related studies and have significant contributions for dataset processing, deep learning structures, imaging techniques and significant contributions to the models that act as a base for this paper research.

The World Health Organization on cancer (2022) gives many international views on colorectal cancer and have detailed its prevalence and the health demanding situations which are connected to it and stressing the significance of early analysis. [1] Nagadia offered the Kvasir dataset on Kaggle, offering annotated endoscopic images which can be treasured for the development and testing of model geared toward detecting gastrointestinal diseases. [2]

Pogorelov et al. (2017) added the authentic Kvasir dataset, a benchmark dataset for multi-class laptop-aided detection of gastrointestinal illnesses that has been widely applied in medical AI research. [3]

Chollet (2015) launched Keras, a high-degree deep gaining knowledge of framework that streamlines the advent and deployment of neural community models. [4]

Kaur et al. (2022) investigated computational intelligence and metaheuristic techniques for detecting brain tumors, emphasizing IoMT-enabled MRI models for medical images assessment. [5]

Mohammed and Thamir (2022) evolved a machine for detecting esophageal cancer using a feed-forward neural network, showcasing wonderful overall performance in category tasks. [6] He et al. (2016) supplied ResNet, a framework for deep residual getting to know that extensively superior images recognition and has become a foundational version for clinical picture classification. [7]

Zhao et al. (2022) examined the use of COMPUTERA-based dimensionality reduction for image classification, demonstrating its effectiveness in improving performance whilst decreasing computational needs. [8]

Schmidhuber (2015) offered a comprehensive evaluation of deep learning techniques, frameworks, and applications, laying the basis for cutting-edge neural networks. [9]

LeCun, Bengio, and Hinton (2015) posted a foundational article on a deep learning framework for classifying gastrointestinal problems. Convolutional Neural Networks (CNNs) have been chosen for his or her strong performance in image popularity, comprising convolutional, pooling, and completely connected layers, culminating in a softmax output for sophistication. To rent present knowledge, transfer studying have grown to be achieved the use of pre-professional model like ResNet50, VGG, and Inception. These networks, which have been professional on huge datasets, have been high-quality-tuned for obligations in medical imaging to enhance characteristic extraction. Furthermore, MobileNetV2 emerge as incorporated to provide a light-weight and inexperienced answer appropriate for real-time use in medical settings. The parameter is used to determine the most effective model by identifying connections and patterns between different properties in a specific input or data set used to train models [5].

III. PROPOSED METHODOLOGY

The proposed methodology is structured into five main sectors to ensure a systematic processing of endoscopic images for gastrointestinal disease prediction using a computer-aided diagnostic machine that utilizes deep learning studying to categories the gastrointestinal diseases that have many issues primarily based on endoscopic images. To overcome these challenges we have a structured process that are divided into 5 key levels: dataset acquisition, preprocessing, model development, training and validation, and performance evaluation. Each degree changed into designed to ensure the accuracy, reliability, flexibility and applicability of the proposed method in scientific settings.

1. Dataset Acquisition

This sector is a data collection process, which incorporates categorized endoscopic image graphs depicting various gastrointestinal diseases, such as ulcers, polyps, gastritis, and esophagitis. Images were sourced from major reachable repositories and diagnosed scientific facts to uphold moral and private rules. To ensure reliability, gastroenterology specialists where the labels are demonstrated in the images and graphs, as inaccuracies in labeling can greatly decrease the performance of the model. The main dataset used in our research is the Kvasir dataset [2, 3]. The dataset became balanced to avoid any bias toward a particular class, ensuring it's far numerous and clinically representative. It was cut up into three subsets: 70% certain for training set, 15% for validation, and 15% for testing. The validation subset assisted in hyper parameter optimization, and the check subset allowed for an independent assessment. This methodical dataset guidance ensured that the model may want to generalize successfully, turning in robust diagnostic skills across numerous GI situations, is the reason allowing dependable clinical application of the model.

2. Preprocessing

In this workflow, preprocessing was needed for improved image quality and dataset consistency. Endoscopic images vary in details, brightness, and noise according to variations in equipment, clinical presentations, or both. Because of such image quality variations, images were first resized to 224x224 pixels by a computer-automated Keras data generator to prevent any unwanted distortion [4]. The images were also denoised using Gaussian and median filters, which improved contrast. Data augmentation was needed for infusion of variety into the set. This step prevented overfitting and simulated real-world variations, revealing crucial mucosal patterns saliently. Data augmentation techniques involved random flips, rotation, zooms, brightness shift, and contrast changes.. The preprocessing pipeline supplied uniform and incredible inputs that progressed feature extraction and overall version accuracy. By organizing a standardized and varied dataset, preprocessing more advantageous on robustness, adaptability, and scientific relevance of the deep getting to know-based on the diagnostic model. Preprocessing became vital for boosting image pleasant and preserving the consistency during the dataset, improve clarity in the model to have high accuracy on during the dataset preprocessing.

3. Model Development

The recognition of model improvement come to be on growing on a study of deep analyzing framework for classifying gastrointestinal problems. Convolutional Neural Networks (CNNs) have been chosen for his or her strong performance in image popularity, comprising convolutional, pooling, and completely connected layers, culminating in a softmax output for sophistication. To rent present knowledge, transfer studying have grown to be achieved the use of pre-professional model like ResNet50, VGG, and Inception. These networks, which have been professional on huge datasets, have been high-quality-tuned for obligations in medical imaging to enhance characteristic extraction. Furthermore, MobileNetV2 emerge as incorporated to provide a light-weight and inexperienced answer appropriate for real-time use in medical settings. The parameter is used to determine the most effective model by identifying connections and patterns between different properties in a specific input or data set used to train models [5].

4. Training and Validation

The version established high accuracy and generalization because of thorough education and validation techniques. The dataset changed into divided into three subsets: 70% for education, 15% for validation, and 15% for sorting out. A grid search modified into hired to great-tune hyper parameters along with analyzing charge, batch duration, and the variety of epochs. When training a data is effectively trained, the results will be a favorable outcomes for the test data [6]. Training became improved on GPU-enabled structures, in particular for complex architectures like ResNet50 and MobileNetV2. After each epoch, validation became done to tune the studying manner and remarkable-track parameters for that reason. Finally, the model validation is evaluated and measured through many metrics and unseen test data such as accuracy, recall, precision and F1-score. In the give up, the model performs category and prediction, offering outcomes that sign the unique form of GI abnormality, like polyps. These model undergoes training and validation to enhance their overall performance. In the give up, the model performs category and prediction, offering outcomes that sign the unique form of GI abnormality, like polyps. This dependent approach ensures that diagnostic predictions are unique, unbiased, and may be without problems replicated, as illustrated in Figure 1.

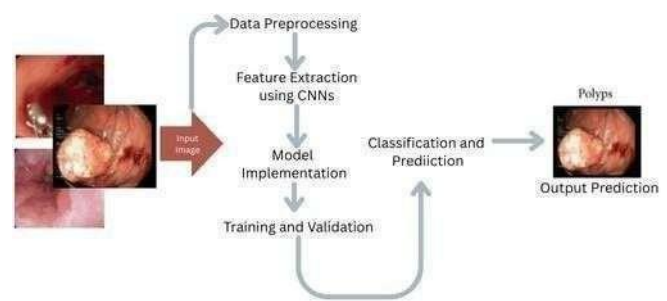


Figure 1. Workflow/Architecture of proposed gastrointestinal diseases prediction system

The proposed structure outlines an in depth deep mastering framework geared in the direction of the automated magnificence of gastrointestinal diseases use the endoscopic image graphs, with the reason of minimizing diagnostic errors and enhancing clinical selection-making. The selected processes encompass information accumulating, preprocessing, and design of model structure, schooling, and thorough evaluation to assure the advent of a sturdy and extensively applicable deep mastering model [11]. To mitigate overfitting and hold balance, diverse records augmentation techniques have been employed, which includes transposing, flipping, rotating, adjusting brightness, blurring, distortion, contrasting, and constrained the most progressed way that emits the techniques to the several most,

adaptive histogram equalization. Following the data augmentation, the overall number of images reached forty thousand. Recently, convolutional neural networks (CNNs) have attained notable progress in quite a few laptop imaginative and prescient programs [9, 10]. Our approach harnesses a ResNet-50 convolutional neural network in constructing an advanced predictive system for gastrointestinal disorders by examining endoscopic images. The main goal of this project was to go beyond basic polyp detection, CAdE, to a real-time diagnostic capability, CAdx, and risk assessment. First, the ResNet-50 is fine-tuned to act as an effective "optical biopsy" tool: it is trained only on images that have been carefully annotated for true pathology, enabling the accurate prediction of the histology of individual polyps, thus distinguishing between benign, or hyperplastic, and pre-malignant, or adenomatous, lesions. These results directly support clinical "resect and discard" strategies that help decrease costs and save time. These predictions are further supported through the integration of Grad-CAM explain ability, which produces heat maps showing the exact visual features the model relies on for its classification decisions.

IV. ALGORITHM

The algorithm we propose is a ResNet50-based Convolutional Neural Network for automatically categorizing gastrointestinal diseases using endoscopic images. It aims to improve diagnostic accuracy and provide reliable decision support for clinicians. ResNet50 is very effective at extracting complex features in networks, which boosts the accuracy of distinguishing between various GI conditions, including polyps, gastric ulcers, and cancer. An interest mechanism focuses on large areas while reducing the effect of the historical limitations of ResNet50. It uses shortcut connections to link different layers. The output from an earlier layer is combined with the input of the following layers [7]. Training is directed by specific bypass- entropy loss, and universal overall performance is classed on unseen facts using metrics like accuracy, precision, recall and F1-score.

Steps for Processing:

1. Start
2. Collect the Dataset from the sources (Endoscopic Images from Kvasir)
3. Image processing,
 - Resize it to a fixed dimension size $m \times n$.
 - Apply augmentation and normalization.
4. Initialize the CNN model (e.g., ResNet50, VGG16).
5. Replace the final classification layer with softmax for GI disease into classes.
6. Make the dataset split into training and testing sets.
7. Train the CNN model on training data:
 - Forward $\rightarrow$ to extract the features.
 - Backpropagation $\rightarrow$ to update the weights.
8. To the validate model on test data.
9. Predict the disease class for the input image.
10. Evaluate the performance of the mode using metrics such as Accuracy, Precision, Recall, and F1-score.
11. End

$$CNN(I) = f(W \times I + B) \quad (1)$$

Where I is the input image, W stands for the weights of the filters, $+$ is the unfairness, $*$ denotes the convolution operation, and f is a non- linear activation feature and in the end B is a Bias term which is after the convolution and then the performance evaluation was computed. The prediction version's efficiency and this formulation has a model on using metrics which evaluate and dependability had been cautiously evaluated.

1. MODELS

The main use of endoscopic images to achieve the unique and reliable class of gastrointestinal (GI) disease by the model, the proposed model is assumed in the correct way that the ongoing neural network (CNN) - model uses VGG16, ResNet50 and Mobilenetv2. Each model has great benefits in composition, general performance and illustration, which specifically uses them for clinical image evaluation. VGG16 is a traditional deep CNN consisting of 16 layers, identified for its consistent shape that uses small 3×3 conversion filters. Although it has large amounts of parameters, it improves images to get difficult spatial information in the images, it is actually necessary to detect micro gastrointestinal abnormalities where minor texture changes can also form the disease. The ResNet50 has remaining connections, besides referred to as a fit connection to solve the faded shield problem in the deep network. It allows intensive learning at the same time to ensure stability under the education system. The ResNet50 is quite effective in extracting complex hiezen skills in networks, which increases the accuracy of differences in many GI conditions that include polyps, gastric ulcer and cancer. MobileNet2 prefers performance using deep different conversions from deep remaining and linear bottlenecks. This progression reduces the amount of parameters at the same time as calculation requirements and the same accuracy are preserved. Mobileentv2 is the best for each light-vest and concise, which is best for a scientific environment in real time, with short and resource-green predictions. By integrating three models, suggestions balanced system capable of delivering accurate, unbiased, and repeatable predictions for computer-aided diagnosis of GI diseases from endoscopic image.

V. EXPERIMENTATION AND RESULT

The proposed CNN size, which is honestly grounded in the reset50 and has the capacity of a mechanism of interest, which has become a test using a dataset of gastrointestinal (GI) endoscopic Snap shot. During the experimental phase, the dataset cuts at final training, cuts to verification and tries to equip the equipment to ensure an objective assessment of the model's efficiency. Pretreatment steps, related to form change, normalization and increase, helped to reduce and increase the normalization of the model. The proposed approach gained a favorite accuracy of 84.5%, in the brain of 87.6%, and a well - balanced sensitivity and forecasts total performance 80%. These conclusions recommend that the version skills stand out on a minimal abortion rate between severe GI problems. The joint of the eye system proved important, as it allowed the model to focus on clinically related areas in the endoscopic images, which improved the clinical capacity of the CNN ridge. Compared to Baseline models such as VGG16 and MobileNetv2, the Reset50- based completely technology showed high performance in sentences of frequent class accuracy and reliability. In summary, experimental conclusions believe that the proposed version can act as a reliable computer-aided diagnostic tool, which can help health experts in the early identification and sophistication of gastrointestinal conditions. Comparing all the three models, the more efficient and more accurate model is ResNet50. So, we are using the ResNet50 model to predict the GI diseases. The models accuracy is illustrated in Table 1.

Model	Accuracy (%)
MobileNetV2	80.5%
ResNet50	84.5%
VVG16	70.1%

Table 1. Performance Metrics

Here comparing various spine architecture such as VGG16 and MobilenetV2, the ResNet50 framework improved them. Although VGG16 is effective to identify the functions at the following level, the absence of remaining connections disrupts its ability to effectively train deep layers, resulting in a lower classification accuracy than the

ResNet50. Contrary to this, the MobileNetV2 provides calculation efficiency due to the mild design, but this efficiency compromises its accuracy, as it seems challenging to capture the fine days of details required to separate micro differences in GI diseases. The main advantage of the ResNet50 is its remaining learning structure. This uses jump connections to deal with the faded shield problem so that the model can learn more complex and detailed functional representations. In addition, the implementation of the attention mechanism improves its efficiency by emphasizing the most clinically related parts of the image and reducing which are not to related background elements. The findings from the experiments display that the cautioned technique outperforms both VGG16 and MobileNetV2 while classifying gastrointestinal diseases across 50 terms. Additionally, the incorporation of meditation techniques in conjunction with schooling more desirable scientific accuracy, positioning the model as a valuable machine for sufferers to discover early signs and symptoms of illness and help doctors in their scientific choice-making.

1. Performance and Evaluation

The mean overall performance studied the reliability of the version in analysis and its suitability for clinical application. Given the increased dimensionality of the features, the CNN feature maps were analyzed with the PCA method [8]. The evaluation measures safeguarded accuracy to degree popular correctness, in addition to precision, keep in mind, and F1-score to measure common place performance specific for each beauty. These measures were particularly significant for unbalanced datasets where too many high-quality gastrointestinal (GI) issues had fewer examples. A confusion matrix have been developed to illustrate the correct classifications toward the expected ones, which helped identify typical misclassifications. Here accuracy play a major role where it measures the predictions more perfectly overall. The performance and evaluation is calculated by the metrics such as accuracy, recall, precision. To evaluate the efficiency of the proposed models, standard classification matrix is used as accuracy, accurate, recall and F1 score. Although this is informative, it cannot represent performance in terms of class imbalance, which is often found in therapy datasets. Accuracy measures the excerpts of real positive predictions in relation to all positive predictions, which highlights the system's ability to reduce false alarms. High precision is important to prevent unnecessary medical procedures. Memory (sensitivity) considers the relationship between real positive cases that the model recognizes. In medical diagnostics, it is important to achieve high recall to ensure that no cases of any diseases are indefinite. Together, they offer intensive evaluation of the Matrix model, which addresses the need to reduce the diagnosis that is left by reducing false alerts, which is important for developing a reliable computer system (CAD) for gastrointestinal diseases.

The formulas to calculate the metrics are,

$$Accuracy = \frac{No. of correct Prediction}{Total No. of prediction} \quad (2)$$

$$Precision = \frac{TP}{TP + FP} \quad (3)$$

$$Recall = \frac{TP}{TP + FN} \quad (4)$$

It's critical to well known that the current studies has its barriers. While the dataset is right enough for checking out functions, it may now not actually constitute the range located in actual

medical eventualities. The structured generalization, future research need to have an awareness on increasing the dataset via incorporating contributions from multiple centers. Compared accuracy graph between the three models have been illustrated in Figure2.

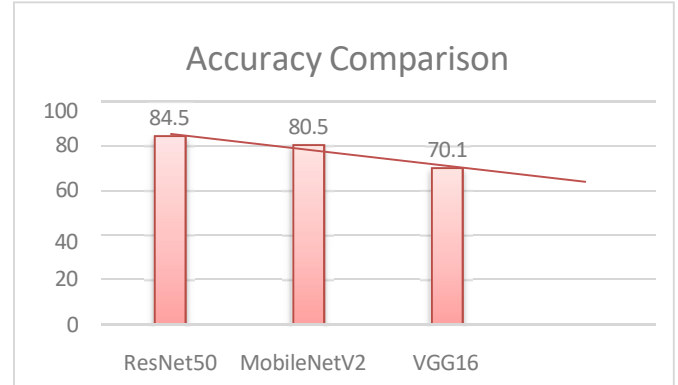


Figure 2. Test Accuracy Comparison

This comparative analysis shows that ResNet50 achieves the highest accuracy due to its deep residual learning, while VGG16 provides strong feature extraction but with higher computational cost. MobileNetV2 is slightly lower in accuracy, ResNet50 offers the best efficiency, making it suitable for real-time clinical use.

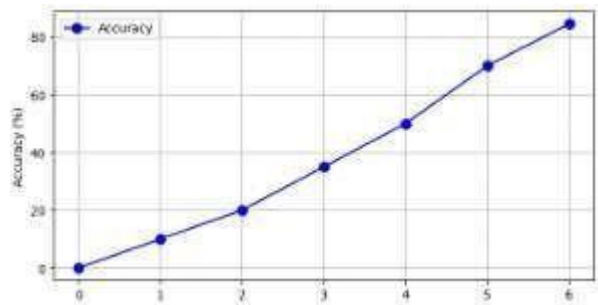


Figure 3. Accuracy graph

VI. CONCLUSION

In this study, we developed a certain neural network using the ResNet50 architecture, expanded with an attention mechanism, especially designed for automatic classification of gastrointestinal (GI) diseases using endoscopic images. The model effectively utilized the remaining learning functions effectively to capture both basic and advanced image functions, while the meditation mechanism enabled the network to focus on clinically important areas. Experimental findings indicated that the proposed approach had an accuracy of 84.5%, a memorial of 87.6% and an F1 point of 85.8%, which crosses traditional CNN models such as VGG16 and MobileNetv2. These results suggest that the ResNet50-based framework provides reliable and strong equipment to help doctors in the initial diagnosis and informed decision-making. Although the results of the proposed method are promising, there are still areas of improvement. Future improvement may include a wide range of clinical data sources to increase the normalization and understanding of the dataset with excessive amounts of samples. To mitigate overfitting and hold balance, diverse records augmentation techniques have been employed, which includes transposing, flipping, rotating, adjusting brightness, blurring, distortion, contrasting, and constrained adaptive histogram equalization. Following the data augmentation, the overall number of images reached forty thousand. In the second step, the prepared records were used for all experiments regarding the model. In the third step, sickness detection and type were achieved with the assistance of interest mechanisms. Feature extraction employs convolutional neural networks (CNNs) to hit upon critical spatial features. The derived functions are then inserted into version implementation, which makes use of state-of-the-art deep learning

frameworks along with ResNet50, MobileNetV2, and VGG16 In addition, integration of advanced interpretable AI techniques can increase openness by highlighting visual signals used in the decision -making process, which promotes trust among health professionals. In addition, detecting a hybrid architecture that combines the recanet50 with a transformer – based model can improve further performance in medical image analysis. The proposed structure provides a strong base for computer aided diagnosis of GI diseases and has an important potential. a fit connection to solve the faded shield problem in the deep network. It allows intensive .The proposed model, which combines remaining and learning with attention mechanisms, showed reliable performance in the classification of gastrointestinal diseases. The findings validated their ability to identify subtle differences in endoscopic images. Although the current accuracy levels are promising, but there is an opportunity for enrichment with large and more different data sets. Inclusion of a clear AI will increase the transparency of the system and increase its clinical acceptance. Refining the contour for real - time use can expand the use in hospital settings. Hybrid deep learning models can improve future progression and clinical accuracy. In summary, this model creates a strong base to create a computer supported tool that supports gastroenterologist and patient results in medical sectors.

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